BT-201 (GS)

B.Tech., I & II Semester Examination, June 2026

Grading System (GS)

ENGINEERING PHYSICS

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Partial — source me Q1 aur Q5–Q8 mile (Q2–Q4 page capture nahi hua)

1. (a) Explain the Particle in a One-Dimensional Box and derive the expressions for the energy eigen values and eigen functions. **7**

(b) Derive the Time-Independent Schrödinger Equation. **7**

2. (source me ye question available nahi tha — original paper dekhein) **—**

3. (source me ye question available nahi tha — original paper dekhein) **—**

4. (source me ye question available nahi tha — original paper dekhein) **—**

5. (a) Define Fermi Energy and Fermi Level. Explain the Fermi level in intrinsic and extrinsic semiconductors. **7**

(b) Explain the construction, working and applications of the CO_2 laser. **7**

6. (a) Discuss the properties of laser beams: monochromaticity, coherence, directionality and brightness. **7**

(b) Derive the expression for the Acceptance Angle and Numerical Aperture (NA) of an optical fiber. **7**

7. (a) State and explain Gauss's Divergence Theorem with its applications. **7**

(b) Calculate the electric field and electrostatic potential due to a continuous charge distribution. **7**

8. Write short notes on any two: **14**

(a) Derive the Continuity Equation (b) V-number (c) Kronig–Penney Model (d) Grating equation.

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